

Welcome!

Applied programming (LIS050)

Robert Östling `robert@ling.su.se`

Maria Skeppstedt `maria.skeppstedt@ling.su.se`

Department of Linguistics
Stockholm University

2026-01-20



Stockholm
University

Purpose of this course in the program

To make you proficient enough in programming to solve real problems in the other courses and your thesis.



What more is there to know?

Major parts of the course:

- Efficient algorithms and program optimization
- Various advanced Python concepts
- Vector processing and the basics of machine learning



Course overview

- We start where we left off in Introduction to programming (LIN020)
- Same overall structure: morning lectures, afternoon lab work
- One written exam (March 17)
- Four lab assignments:
 - Lab 1: efficient programming (starts now!)
 - Lab 4: solving an NLP problem (runs over the entire course)
 - Lab 2: working with NumPy and vector representations
 - Lab 3: machine learning with pyTorch
- This is a Python course, but we sneak in as much NLP as we can to prepare you for later courses



- We start out with two groups of at most 20 students.
- We are unsure about the level of participation – who among you plan to attend non-mandatory classes regularly?
- If both groups have labs in parallel, you can just pick the classroom with seats left.
- Otherwise, please coordinate among yourselves.



Peer code review

- We make extensive use of peer code review in this class. You will write a lot of code, teachers can't comment on everything.
- Peer review takes place at Språkstudion, some of the Tuesdays, either 13:00-15:00 or 15:00-17:00.
- You will work in pairs, and student B will review student A's code as follows:
 - ① Student A demonstrates their program, shows that the output is correct, etc.
 - ② Student B looks at the code quality and asks questions.
 - ③ Student B fills out a brief checklist on Athena.
 - ④ Student A updates the code according to comments from student B
- When student A is happy with the updates, the student submits the code in Athena. Then the student presents the code to a teacher.
- The review process is a great opportunity to learn.
- You may also ask an LLM to review your code at any time.

Note: you will need to explain the details of your code to the teacher, so you need to understand every part of it!



- While it is not allowed to have someone (or something) else write code for you, we do encourage cooperation.
- Things to pay attention to:
 - Lab assignments should be solved by **you**. You are expected to be able to explain your code to the teacher and/or classmates.
 - During the exam, follow the rules given and instructions from the exam supervisor.
- If you do the above, you will be fine.



Grading

- To pass this course, you need to **both**:
 - ① The pass the exam
 - ② To pass **all** four lab assignments. You do that by:
 - ① For lab 1–3: review the code of a peer, submit your code in Athena and thereafter present your code to a teacher.
 - ② For lab 4: Present your program to the class. Submit your code in Athena.
- The course is graded on a scale from A (best) to F (failed)
- The exam is graded as pass/fail
- Each lab assignment is graded as failed or passed with/without distinction
- The number of “pass with distinction” determines your grade:

#distinctions	grade
0	E
1	D
2	C
3	B
4	A



Obtaining help

- We do not have “office hours” as such
- Instead, we spend a lot of time in the classroom, with time for individual questions
- For administrative or sensitive matters that you do not want to discuss in class, you can email robert@ling.su.se or maria.skeppstedt@ling.su.se
- For help with course contents:
 - Ask the teachers during class
 - Ask your fellow students
 - Ask StackOverflow, ChatGPT, forums, etc.



- Python book: *Think Python* (a few selections)
- Documentation and tutorials for various Python libraries
- Jurafsky & Martin: *Speech and Language Processing* (more in the NLP course)
- Goodfellow et al.: *Deep Learning* (a few selections)
- A couple of papers on sustainable computing
- More details in the course description on Athena

